

CHAPTER 17

From Cavity Polaritons to Photonic Crystals

1. The concept of a Fabry–Perot resonator in the form of a (micro) cavity
2. The cavity polaritons as a mixed state between a resonance in a solid and a cavity resonance.
3. Photonic band gap materials.

17.1 Cavity Polaritons

17.1.1 The Empty Resonator

Fabry–Perot resonator

➤ The case of plane-parallel surfaces:

- An incident plane, monochromatic wave will be partly reflected and partly transmitted at each of the two mirrors.
- In the case:

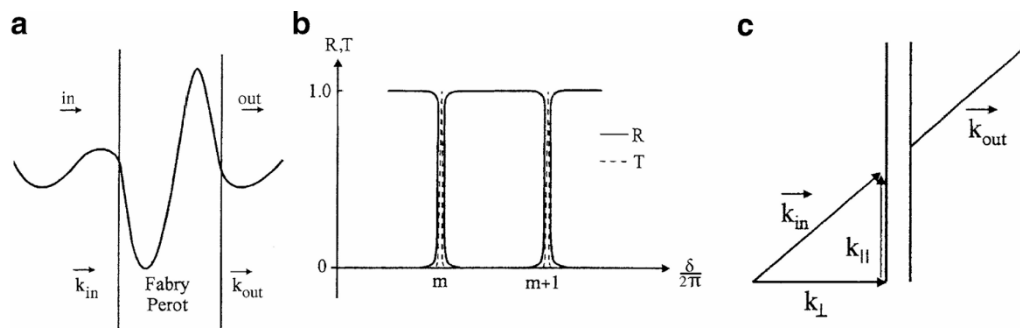
$$m \frac{\lambda}{2} = d \quad \text{or} \quad k_{\perp} = m \frac{\pi}{d} \quad m = 1, 2, 3, \dots$$

all partial waves in the resonator interfere constructively

- For the normal component of the oblique incident:

$$k_{\perp, m} = m \frac{\pi}{d}; \quad m = 1, 2, 3, \dots \quad \Delta k_{\perp} = k_{\perp, m+1} - k_{\perp, m} = \frac{\pi}{d}$$

$$\mathbf{k} = \mathbf{k}_p + \mathbf{k}_{\perp}$$



(a) Schematic drawings of the incident, internal and transmitted fields in a Fabry Perot resonator for normal incidence. (b) The spectra of total transmission T and reflection R as a function of the phase shift per roundtrip. (c) The decomposition of the incident and transmitted wave vectors for oblique incidence

■ 17.1.2 Cavity Polaritons

- We bring some matter into the resonator, connected with some absorption.
- Fig. 17.2a: the dispersion relation of a resonator mode $\hbar\omega_{FP}(k_p)$ and of an excitation in matter.
- Fig. 17.2b: luminescence spectra obtained under band-to-band excitation as a function of the angle of observation Θ relative to the normal of the cavity.
- Fig. 17.2c,d: the layout of the sample and the dispersion relations deduced from the luminescence data.
- If the resonator is made wider and wider.
- If the resonator is, too narrow.
- A prerequisite for cavity polaritons:
The dephasing time of the excitation in matter T_2 must be longer than the roundtrip time of the resonator.

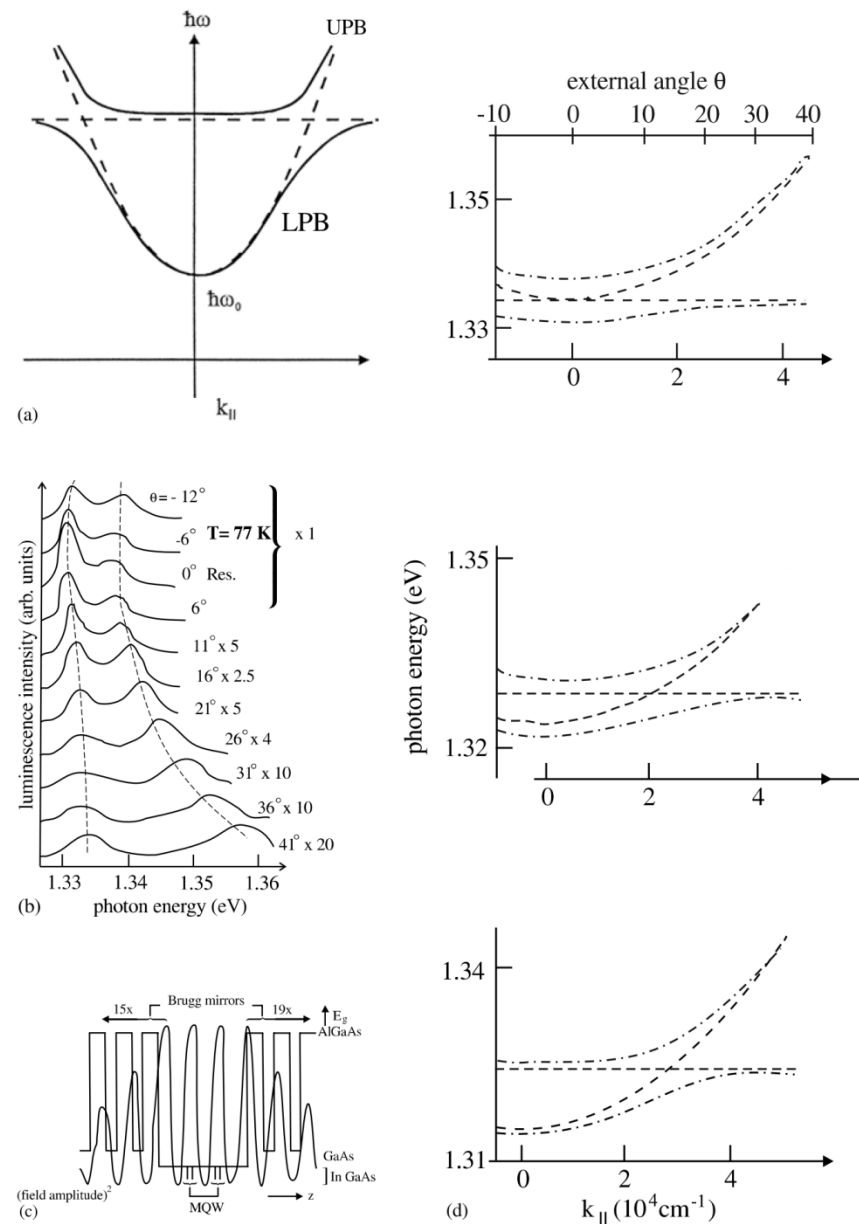


Fig. 17.2. (a) Schematic drawing of the dispersion relation of a microcavity and of an exciton resonance without (dashed line) and with coupling (solid line) LPB and UPB stand for lower and upper polariton branch. Compare to Chaps. 5, 13 and 15. (b) Experimental luminescence spectra for various external angles of incidence of the structure shown in (c) and the measured and calculated data with coupling (-----) and the calculated dispersion without coupling (- - -). According to [94H1]

17.2 Photonic Crystals and Photonic Band Gap Structures

17.2.1 Introduction to the Basic Concepts

A photonic band gap material: in analogy to the band gap for electrons in crystals a material, in a certain energy interval no propagating light modes.

$$d_i = \frac{\lambda_i}{4} = \frac{c\pi}{2n_i\omega_0} \quad i = \text{I, II}$$

Fig. 17.2d :a photonic energy gap.

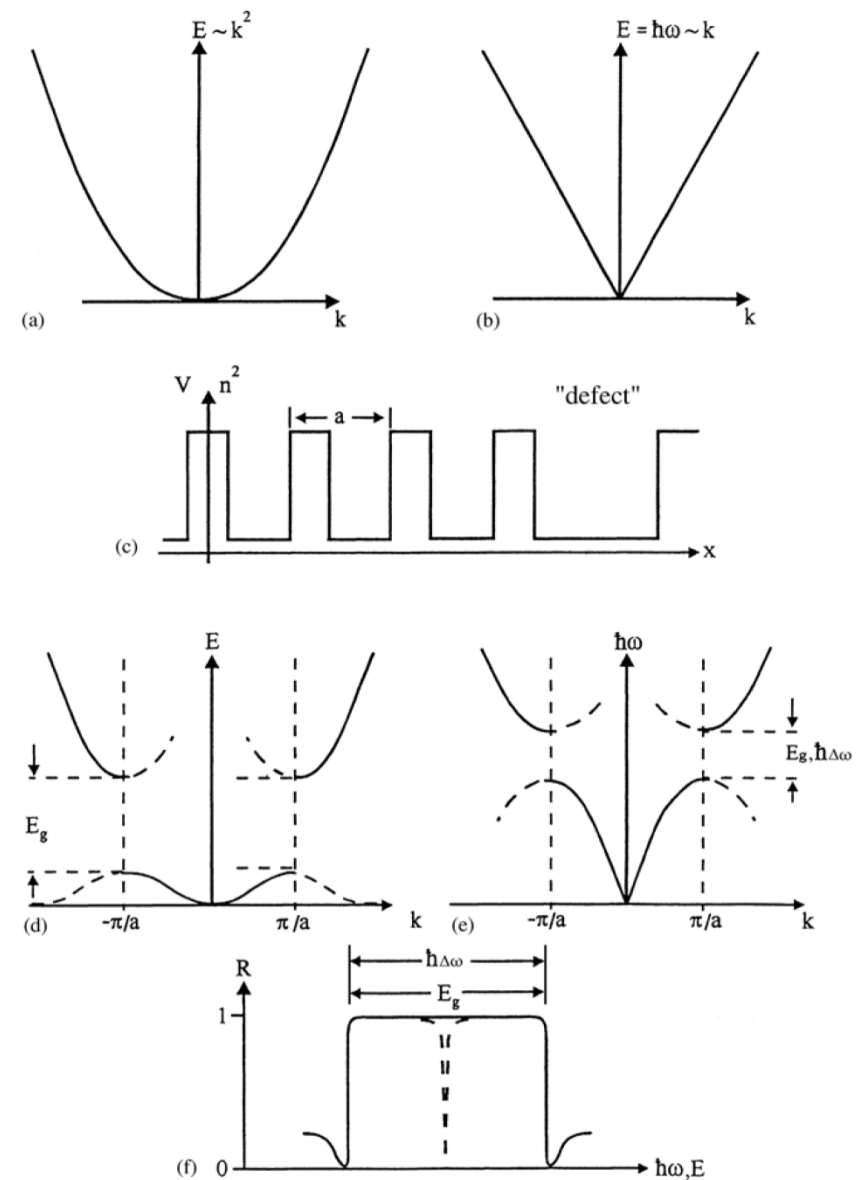


Fig. 17.3. (a) The parabolic dispersion of electrons in a constant potential V_0 . (b) The linear dispersion of photon-like polaritons in a material with constant refractive index n_0 . (c) The periodic potential of the Kronig-Penney model with a “defect”. (d,e) The resulting dispersion relations and band gaps for electrons and light without defects. (f) The reflectivity of a Bragg mirror and of a Fabry-Perot cavity formed by two Bragg mirrors with a cavity (solid and dashed lines, respectively). All drawings are schematic according to [01K1]

■ 17.2.2 Realization of Photonic Crystals and Applications

◆ **Two-dimensional structures:** (Fig.17.4) periodic (hexagonal) arrays of holes in Si that have a spacing constant on the order of $1\mu\text{m}$ and some are $10\mu\text{m}$ deep.

➤ This structure has a true two-dimensional photonic band gap:

$$\omega_0 \approx 0.4c \frac{2\pi}{a} \quad (17.13)$$

➤ Obviously, the band structure depends on the polarization of the light. And the gap is narrower for the electric field polarized parallel to the tubes than for the orthogonal polarization.

➤ Obviously a has to be chosen such that $\hbar\omega_0$ falls in the transparent region of Si.

➤ A chain of defects, missing holes, act for frequencies in the photonic gap as a wave guide. (Fig.17.4)

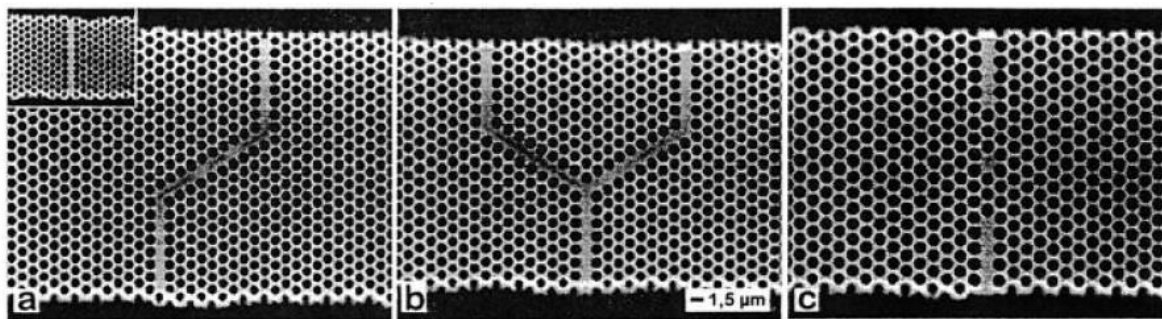


Fig. 17.4. (a) A hexagonal periodic array of cylindrical holes in Si with a wave-guide (b) a beam splitter (c) a resonator [98B1,99B2]

- The application potential of such structures:
 - The integration of wave guides with dimensions $\sim \mu\text{m}$ on a Si chip.
 - **Drawback:** high insertion losses and much larger than the building blocs in highly integrated electronic chips.

◆ Three-dimensional structures:

- Different lattices and ways to create them have been proposed, i.e. Yablonovite
- Natural opal, has gaps for certain directions in k -space.
- The idea of an inverted opal is now to fill or impregnate the voids between the SiO_2 spheres by Si. By selective etching to remove SiO_2 resulting in a periodic arrangement of air-spheres surrounded by Si.

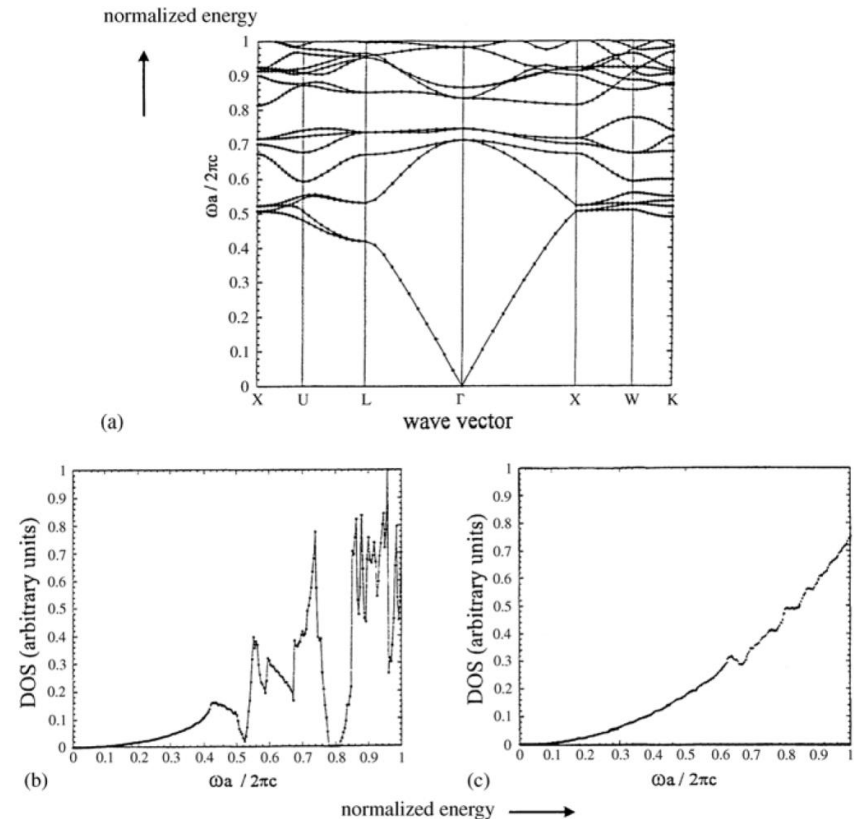


Fig. 17.5. (a) The calculated photonic band-structure of close-packed air spheres in Si, a so-called inverted opal. (b) The resulting density of states (DOS), showing clearly a gap around $\omega a / 2\pi c \approx 0.8$ for this system and for (c) SiO_2 spheres in air corresponding to natural opal [99J1]

硅光子学

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谢谢大家!

张新海

13510648248

Email: zhangxh@sustech.edu.cn

工学院南楼341

